

Lab report draft: Gamma radiation and attenuation

Introduction

This experiment aims to investigate the interaction between gamma radiation and matter, specifically focusing on the attenuation of gamma rays at different energies as they pass through lead. By analyzing the energy spectrum of gamma radiation from Cs-137 and Co-60, and measuring the corresponding attenuation coefficients, we aim to quantify how effectively lead absorbs gamma radiation across varying energy levels. This will provide insight into the processes that govern gamma radiation interactions with matter, such as photoelectric absorption and Compton scattering, and how these interactions are influenced by factors like photon energy and material thickness.

Theory

Gamma radiation

Gamma radiation or gamma rays are a form of electromagnetic radiation, or photons, typically with energies ranging from 10 keV to several MeV. They are created during the de-excitation of an atomic nucleus, and their energy depends on the excited state and the specific nuclide. These energies are generally higher than those of characteristic X-rays, which are produced when electrons transition to lower energy states by emitting photons, usually falling between 1 keV to 100 keV. It's important to note that the distinction between X-rays and gamma radiation is not strictly based on energy—since their energy ranges can overlap—but rather on their origin: X-rays arise from electron transitions in atoms, while gamma rays result from nuclear de-excitation.

While nuclei can become excited through several different processes, they are commonly produced as a result of radioactive processes such as alpha and beta decay, in which the daughter nuclide can be created in an excited state. Excited nuclei will eventually decay to their lowest possible energy state, often by emitting a gamma photon whose energy corresponds to the difference between the excited and ground states. Alternatively, the nucleus may undergo internal conversion, where the excitation energy is transferred to an atomic electron, ejecting it from the atom. An electron from a higher energy level then de-excites by dropping into the lower energy shell, and in doing so, emits a characteristic X-ray rather than a gamma ray.

One example of a nuclide that can be created in an excited state is barium-137 (Ba-137), which forms through beta decay of cesium-137 (Cs-137). As Cs-137 decays, the resulting Ba-137 has a 5.4% probability of being formed in its stable ground state and 94.6% probability of being formed in an excited state. The energy difference between the stable and the excited state is approximately 662 keV, and when measuring the energy spectrum for Cs-137, it will contain a prominent peak at this energy. Cobalt-60 (Co-60) also undergoes beta decay to form nickel-60 (Ni-60), which primarily emits gamma radiation at 1173 keV and 1332 keV.

The general shape of a gamma spectrum can be divided into several parts, each linked to different interactions between gamma radiation and matter. Photopeaks are sharp and well-defined because they correspond to the complete absorption of the incoming gamma photon's energy, usually through the photoelectric effect. Each photopeak represents the specific energy of gamma photons emitted during nuclear de-excitation.

The Compton continuum is a broad, continuous region of the spectrum that appears before the main photopeak and results from Compton scattering. In this process, a gamma photon transfers only part of its energy to an electron, ejecting the electron while the remaining energy is carried away by the scattered photon. The energy of the scattered photon depends on the scattering angle, leading to a range of energy losses, which creates the continuous nature of this part of the spectrum. The edge of this region, known as the Compton edge, corresponds to the maximum energy transfer possible in Compton scattering and is always a lower energy than the full-energy photopeak.

Additionally, pair production peaks are associated with the creation of a positron-electron pair when a gamma photon has an energy of at least 1.022 MeV (the combined rest mass of an electron and a positron). The positron subsequently annihilates with an electron, emitting two photons of 511 keV each, which can appear as distinct peaks in the spectrum. However, this phenomenon does not occur for Ba-137 gamma emissions, as its gamma photon energy (662 keV) is below the energy threshold for pair production, and is relatively unlikely even at 1332 keV, as characteristic for Ni-60.

Finally, internal conversion is an alternative process to gamma emission. In internal conversion, the nucleus transfers its excess energy directly to one of its atomic electrons, which is then ejected from the atom. This process does not produce a gamma photon. Instead, the vacancy created by the ejected electron is filled by another electron from a higher energy level, resulting in the emission of a characteristic X-ray. These X-rays appear at lower energies in the spectrum and are distinct from the gamma photopeaks and the Compton continuum.

Nal scintillator detector

A NaI scintillation detector works by converting incoming gamma radiation into visible light. When a gamma photon interacts with the sodium iodide (NaI) crystal, it can excite the atomic electrons of the crystal by transferring part or all of its energy, either through the photoelectric effect, Compton scattering, or pair production. Afterward, atoms deexcite by emitting light, referred to as scintillation. The scintillation light is collected by the photomultiplier tube, which amplifies the signal and converts it into an electrical pulse. The size of the pulse corresponds to the energy deposited in the NaI by the incoming radiation, allowing the energy spectrum to be recorded.

Gamma attenuation

As gamma radiation passes through a material, the interactions detailed in the above section causes the photons to transfer some or all of their energy to the material, which results in a decrease in radiation intensity. Despite the complexity of the interactions, the attenuation of gamma radiation in a material can be described by a simple exponential function:

$$I(x) = I_0 e^{-\mu x}$$

where I_0 is the initial intensity of the radiation, x is the distance traveled in the material, and μ is the linear attenuation coefficient, which is both material and energy dependent. It is influenced by the relative probabilities of the three primary interaction processes: photoelectric absorption, Compton scattering, and pair production, each of which dominates at different photon energy ranges.

At low photon energies, the photoelectric effect is the dominant interaction, particularly in materials with higher atomic numbers, such as lead. At intermediate photon energies, Compton scattering becomes the primary interaction and therefore, μ is less dependent on the atomic number and more influenced by the density of the material in this range. At high photon energies, particularly above 1.022 MeV, pair production becomes more significant. In this case, the attenuation coefficient once again becomes more dependent on the atomic number, as heavier atoms are more likely to induce pair production.

The overall attenuation coefficient μ is a combination of the contributions from these three processes, and it varies depending on the energy of the incident gamma photons. In general, higher-energy photons penetrate further into a material because their probability of interaction decreases as energy increases.

Experimental setup

The experiment aimed to investigate the attenuation of gamma radiation by lead using a NaI scintillation detector, calibrated using the known energy peaks of Cs-137 and Co-60. The experiment was carried out in three stages: recording energy spectra of Cs-137 and Co-60, calibrating the detector, and measuring the attenuation coefficient of gamma radiation in lead. The Tukan software was used for data acquisition and analysis, with a collection time set to 120 seconds for each measurement.

First, the energy spectrum of Cs-137 was recorded. A closed Cs-137 sample was placed inside a lead cave. Placed above it was a NaI scintillator connected to a photomultiplier (PMT). The lead cave has a dual purpose; reducing noise in the energy recordings from background radiation and limiting exposure and dosage during the experiment. The spectrum was analyzed in order to identify the expected peak at 662 keV.

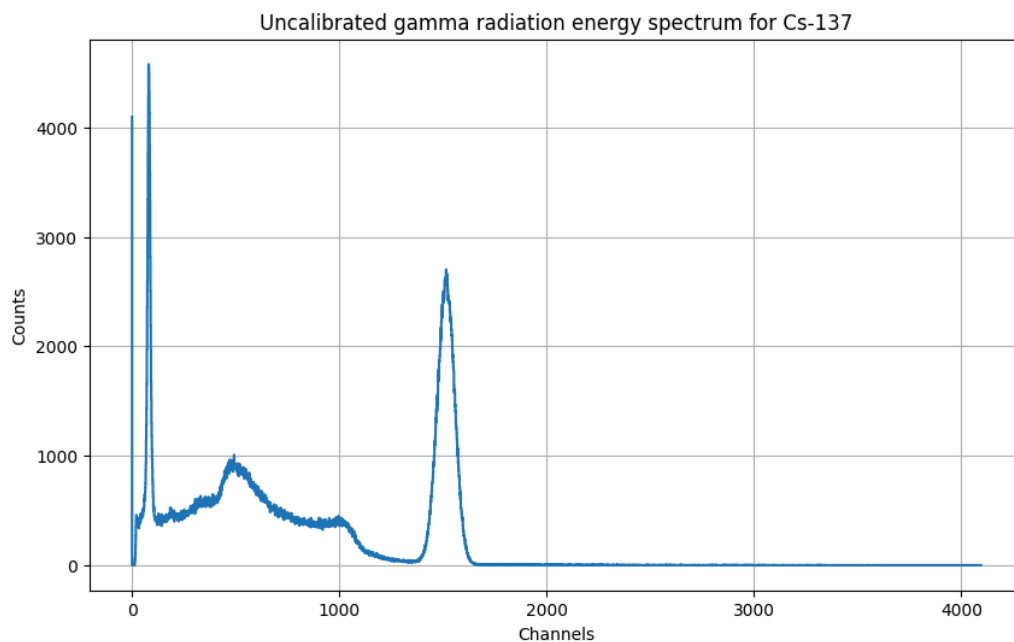
Once identified, a closed Co-60 sample was placed inside the lead cave alongside the Cs-137 sample and the joint spectrum was analyzed. Due to the known decay processes associated with Co-60, two more prominent peaks were expected and identified. Given the three peaks, the NaI scintillator was calibrated using the corresponding known energies to ensure that the detector accurately correlated the channel numbers in the recorded spectra with the corresponding photon energies.

Finally, the gamma radiation attenuation coefficient of lead as a function of energy was studied by inserting lead plates in the lead cave in between the gamma-emitting samples and the NaI scintillator. The intensity was estimated using the area underneath the energy peaks at 662 keV, 1173 keV, and 1332 keV as it corresponds to the total number of recorded

photons associated with those energies. Measurements were taken for varying thicknesses of lead, starting with no lead and continuing with increasing thicknesses in instances of 4 mm. An exponential function was then fitted to the data points in order to obtain the experimental value of the attenuation coefficients for the different energies.

Results

The uncalibrated gamma energy spectrum obtained for the Cs-137 sample without lead shielding is displayed in figure 2. In the spectrum, we can observe a high, narrow peak at the lower energies. At somewhat higher energies, there is a continuous section containing a less pronounced peak in the middle. Following this continuous part, we can observe a prominent peak. These three sections are consistent with the theoretical gamma energy spectrum; characteristic X-rays emitted as a result of internal conversion of the Ba-137 nucleus, Compton scattering of the gamma radiation containing a Compton peak corresponding to the most probable scattering angle in this geometry, and a full-energy photopeak, respectively.



The intensity of the gamma radiation from the Cs-137 and Co-60 samples at varying lead shielding thickness is shown in figure 4 and figure 5.

After measurements, of the peaks surface area (3 different peaks : 1 from 137-Cs and 2 from 60-Co) for different thickness of lead we obtain the following results :

We decided to split the plot of the peak of ^{60}Co and ^{137}Cs because as we saw before on the spectrum without any shieldings, the peak's intensity were much lower for the cobalt's peaks than for the cesium peak. It allows the plots to be more clean and readable when separated.

According to the general tendency of all three curves we decided to plot an exponential regression. For all the energy of the gamma particle, the reliability coefficient R, that states whether our regression seems coherent, is pretty high and close to 1.

Those results fits then really nicely with the theory that states that :

$$I(x)=I_0 \cdot e^{-\mu x}$$

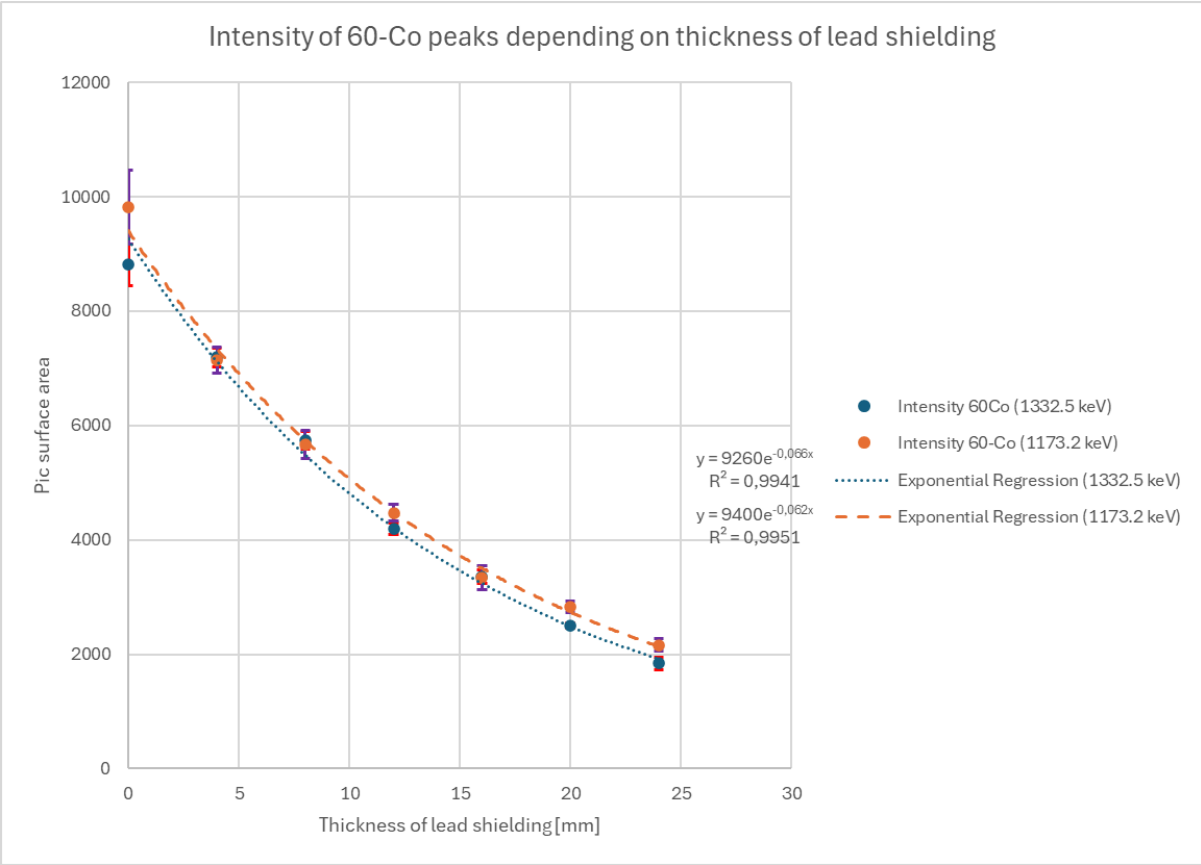
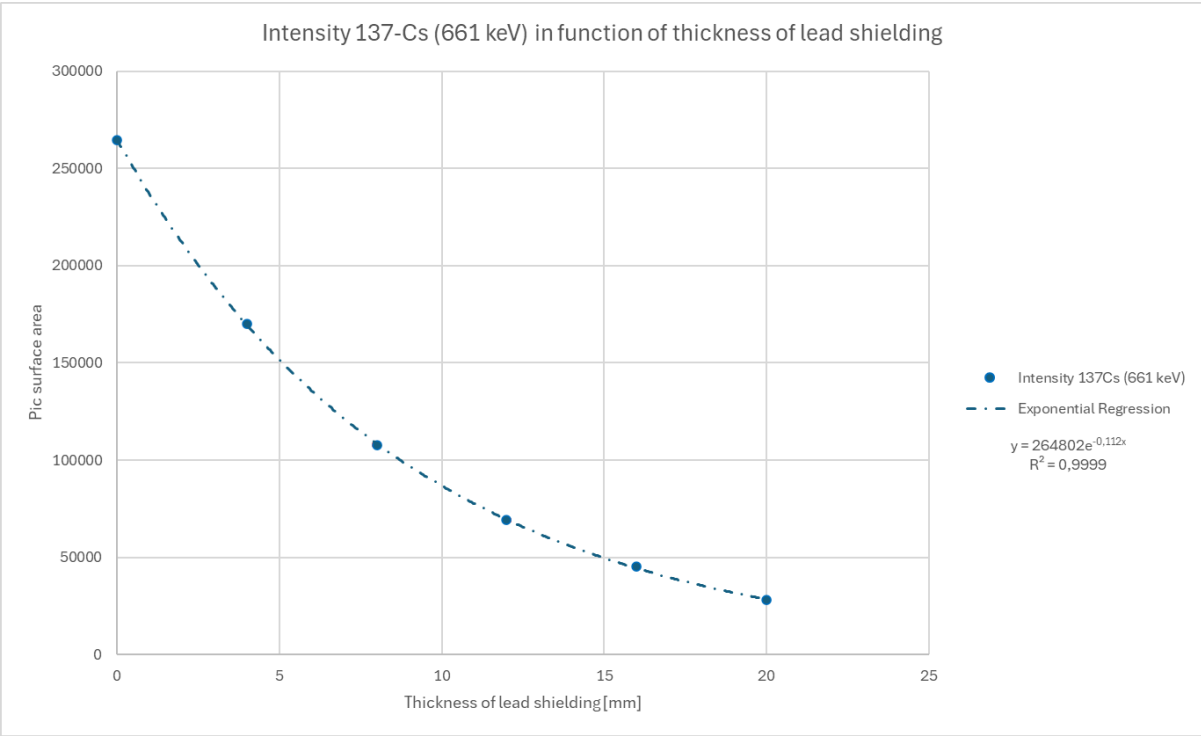
with μ the gamma absorption coefficient and x the distance path in the material (here lead shielding)

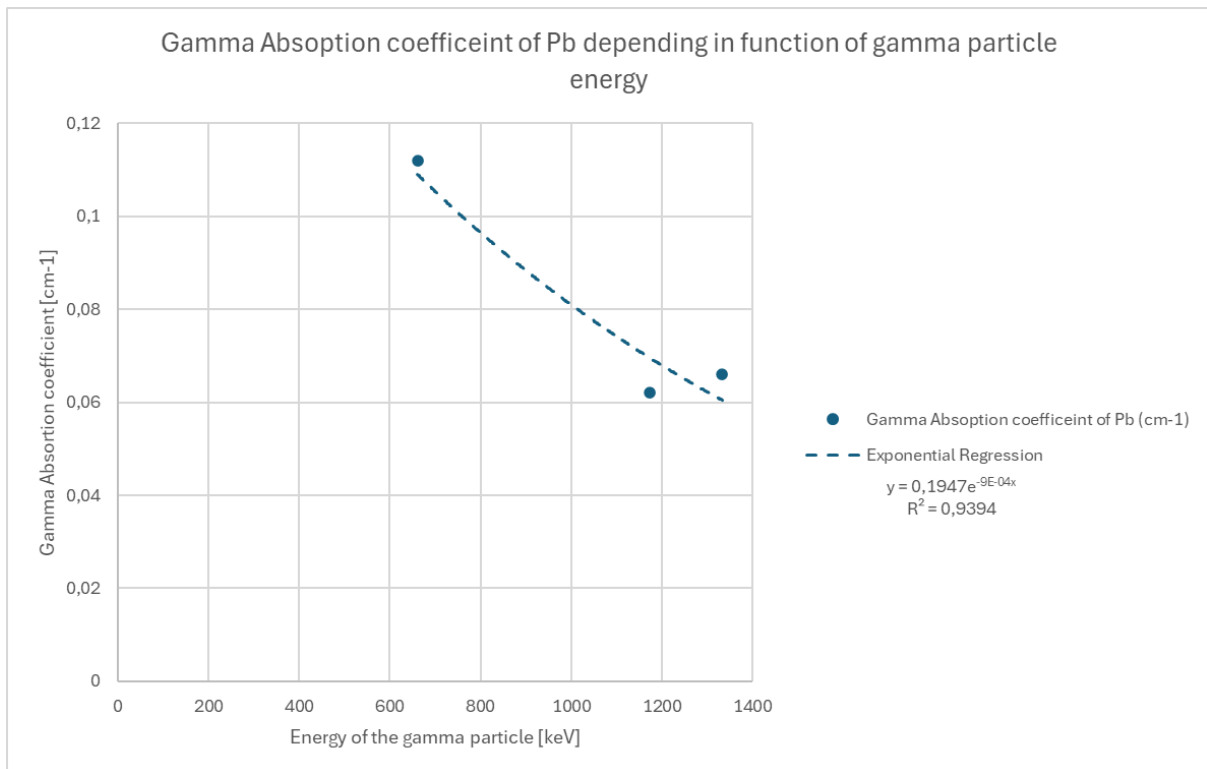
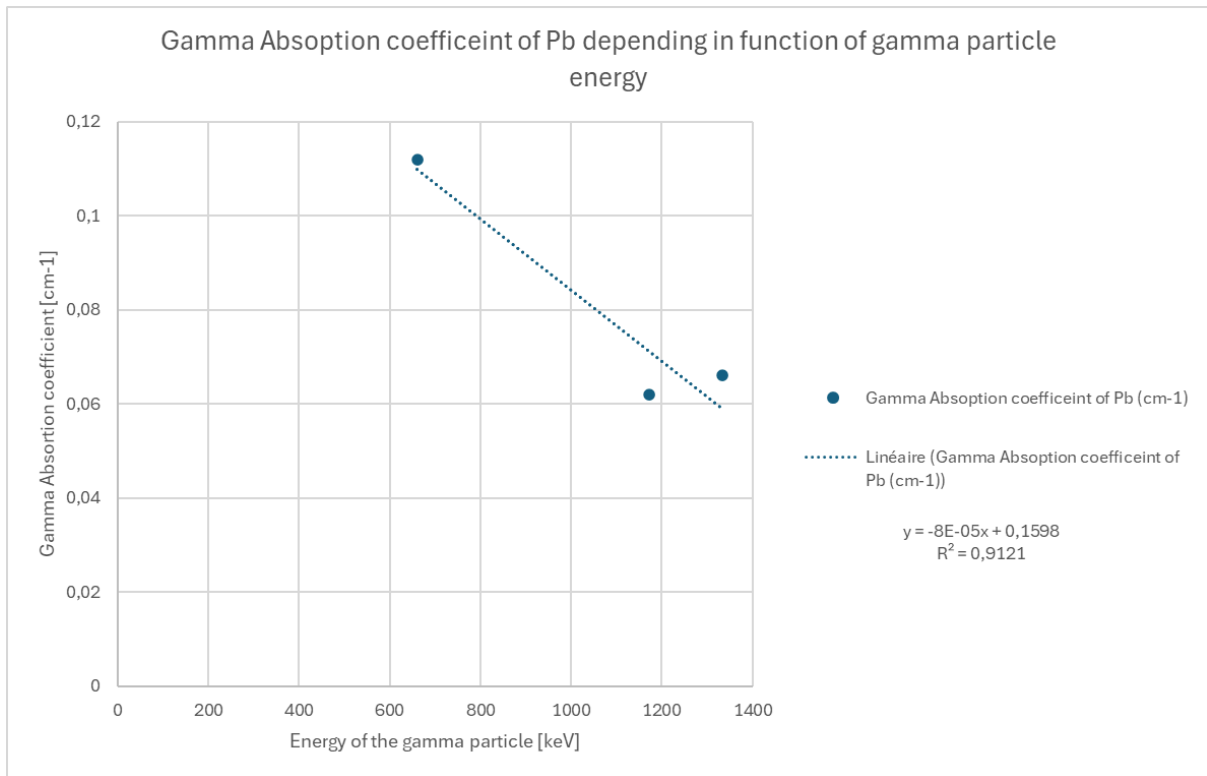
So from those two curves we can extract the exponential regression coefficient for each particle energy and then plot $\mu(E)$ for Pb.

This gives us the following curve :

the energy spectrum

Conclusion





<https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z82.html>